Anomaly Detection for Anti-Money Laundering: Literature Review

Internal research note – January 2026 (v1)

Scope. Survey of ML/DL methods for AML anomaly detection, focusing on 2020–2026. Covers graph-based, deep learning, traditional ML, contextual/KYC, feature engineering, datasets, and emerging paradigms. Not a formal paper—a working reference for project planning.

Audience. Data scientist with anomaly detection experience starting an AML project.

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1 Graph and Network-Based Approaches

Financial transactions form natural graphs (nodes = accounts, edges = transfers). Graph methods detect laundering patterns invisible in tabular data.

1.1 GNNs for AML

The Elliptic dataset (Weber et al., 2019, KDD Workshop) is the foundational benchmark: 203K Bitcoin transactions, 234K edges, ∼2% illicit. GCN achieved ∼0.97 AUC, showing graph structure adds value over node features alone.

Cheng et al. (2024, *Frontiers of Computer Science*, arXiv:2411.05815) survey 100+ GNN methods for financial fraud, finding GNNs substantially outperform traditional approaches.

Method	**Dataset**	**AUC**	**F1 (illicit)**	**Note**
Random Forest	Elliptic	0.96	0.47	Node features only
GCN	Elliptic	0.97	0.52	+graph structure
GAT	Elliptic	0.97	0.51	Attention-based
GraphSAGE	Elliptic	0.97	0.54	Inductive
EvolveGCN-H	Elliptic	0.97	0.58	+temporal
ATGAT	Elliptic++	0.91	–	9.2% over XGBoost
CARE-GNN	YelpChi	0.94	0.86	Anti-camouflage
PC-GNN	YelpChi	0.97	0.88	Imbalanced learning
GraphSAGE	AMLSim	0.85–0.95	0.70–0.85	Varies by pattern

Key architectures:

- **GCN/GAT/GraphSAGE**: Standard baselines. GraphSAGE preferred for inductive (unseen node) settings. Recent temporal-aware GAT (ATGAT) on Elliptic++ achieves 0.913 AUC.
- **Improved HybridNet** (2025): Combines GraphSAGE + GATv2 + temporal encoding.
- **LineMVGNN** (2025): Line-graph multi-view GNN for directed financial graphs.
- **LayerWeighted-GCN**: Adaptive layer weighting across GNN depths.

1.2 Self-Supervised and Contrastive Learning on Graphs

Label scarcity is fundamental in AML (labeled SAR data is rare and sensitive):

- **LaundroGraph** (Cardoso et al., 2022, ECML-PKDD Workshop): Self-supervised graph embeddings → XGBoost. F1 improved ∼5–10% over raw tabular features.
- **Bit-CHetG** (2024): Subgraph contrastive learning on heterogeneous graphs for laundering group detection.
- **HCLNet** (2025): Hypergraph contrastive learning for group-level fraud semantics.
- **Causal-DHG** (2024): Dynamic hypergraph + causal intervention + multi-view contrastive learning.
- **GRAD** (2025, TheWebConf): Guided relation diffusion + supervised graph contrastive learning from limited labels.
- **SpaceGNN** (2025, ICLR): Multi-space projections with distance-aware propagation.

1.3 Heterogeneous Graphs

Real financial networks have multiple entity types (persons, accounts, merchants, devices) and relation types:

- **CARE-GNN** (Dou et al., 2020, CIKM): Camouflage-resistant; RL-based relation selector. AUC 0.94 on YelpChi.
- **PC-GNN** (Liu et al., 2021, WWW): Pick-and-choose sampling for imbalance. AUC 0.97.
- **xFraud** (Rao et al., 2022, VLDB): Industrial-scale heterogeneous GNN deployed at eBay scale with explainability. Billions of nodes.
- **HAN**: Hierarchical attention (node-level + meta-path semantic-level). 2–8% AUC improvement over homogeneous GNNs.
- **HTGNN** (2025): Heterogeneous temporal GNN integrating spatial, temporal, and semantic information for real-time detection.

1.4 Temporal and Dynamic Graphs

- **EvolveGCN** (Pareja et al., 2020, AAAI): RNN evolves GCN weights over time. F1 0.58 vs. 0.52 for static GCN on Elliptic.
- **TGN** (Rossi et al., 2020): Memory-augmented temporal graph network with per-node memory. Adapted for AML by subsequent works.
- **GTCT** (2025): Graph-Temporal Contrastive Transformer for transactional behavior modeling.
- **Continual Graph Learning for AML** (Deprez et al., 2025, *WIREs Computational Statistics*): Reviews replay-based, regularization-based, and architecture-based strategies against catastrophic forgetting.
- Streaming GNNs from Ant Group, PayPal for real-time processing on evolving graphs.

1.5 Subgraph and Community Detection

Money laundering involves coordinated account networks. Node-level detection misses this:

- **FlowScope** (Li et al., 2020, KDD): Dense subgraph detection for scatter-gather laundering patterns. Orders of magnitude faster than subgraph enumeration.
- **SMoTeF** (2024, *Applied Intelligence*): Smurf detection using temporal order and flow analysis. Applied to 180M+ transactions, 31M accounts.
- **GARG-AML** (2025): Scalable, interpretable smurfing detection. Outperforms SOTA on large-scale networks.
- **AMLGaurd** (2025): Unsupervised GAE-based detection via reconstruction discrepancy.
- **Graph Feature Preprocessor** (2024, KDD): Real-time subgraph pattern mining on streaming transactions. Multicore parallelism.
- **The Shape of Money Laundering** (2024): Subgraph representation learning on Elliptic2 revealing laundering scheme shapes (peeling chains, nested services).

2 Deep Learning (Non-Graph)

2.1 Sequence Models (LSTM, GRU)

Transaction sequences are temporal $\Rightarrow$ RNNs are natural:

- LSTM/GRU outperform SimpleRNN (vanishing gradients). GRU often preferred for AML: fewer params, AUC-ROC > 0.7. GRU matches or beats LSTM on most AML benchmarks.
- Attention-enhanced LSTM (Niu et al., 2021): Attention focuses on most suspicious subsequences.
- Jensen and Iosifidis (2023): Replace rule-based triggers with auto-extracted sequential features at bank level.

2.2 Transformers

- **Tang and Liu (2024)**: Unified Transformer-BDI with distributed knowledge distillation. F1 92.87%, AUC 96.73%.
- **TREASURE** (KDD 2026): Transformer foundation model for high-volume transaction understanding.
- **TransactionGPT** (2025): GPT-style pretraining on payment data for behavioral anomaly detection.
- **FraudGT** (2024, ICAIF): Graph Transformer for financial fraud.
- **TabTransformer, TransTab, BERT, RoBERTa** applied to structured transaction data (2024).
- **Spatial-Temporal-Aware Graph Transformer** (2024, IEEE TII).

2.3 Autoencoders and VAEs

Trained on normal transactions, high reconstruction error = anomaly:

- **VAE for CC Anomaly** (2024, *Big Data Mining and Analytics*): CNN-based VAE; hybrid VAE + classifier models.
- **GAN+VAE Hybrid** (2025): Combines GANs and VAEs for large payment flow anomaly detection.
- **LVAESC-SFD** (2026): Correlation-based feature selection + VAE, 98.24% accuracy.
- **Denoising Autoencoder** (Bundesbank, 2023): Detection + interpretability on financial tabular data.
- **Self-Attention + VAE** (2024, *Expert Systems with Applications*): Fund transfer fraud with attention for long-range dependencies.
- Shallow autoencoders (1–2 layers) often sufficient for tabular AML. AUC ∼0.85–0.90 on CC fraud, ∼0.80–0.85 on AML-specific data.

2.4 GANs for AML

Dual purpose: synthetic data generation and direct anomaly detection.

- **WGAN for data augmentation**: Synthetic fraudulent transactions improve F1 by 10–15%.
- **SMOTE-GAN hybrids** (2023): SMOTified-GAN, SMOTE+GAN, GANified-SMOTE variants.
- **Advanced R-GAN** (2024): Regularized GAN with spectral normalization against mode collapse.
- **CTGAN, Copula GAN, TVAE**: State-of-the-art synthetic data tools. 96–99% utility equivalence to production data.
- Alan Turing Institute project (commenced autumn 2024): GAN-based synthetic AML data for UK FCA initiative.

2.5 LLMs and Foundation Models

Newest frontier (2025–2026):

- **FLAG** (KDD 2025): LLM-enhanced GNN for fraud detection.
- **GuARD** (KDD 2025): Text-rich, graph-informed language model for anomaly detection.
- **DGP** (AAAI 2026): Dual-granularity prompting with graph-enhanced LLMs.
- **AuditCopilot** (NeurIPS 2025 Workshop): LLMs for bookkeeping fraud.

- **Co-Investigator AI** (2025): Agentic framework for faster, more accurate SAR generation.
- **FinGPT** (IJCAI 2023): Open-source financial LLM.

3 Traditional ML

3.1 Gradient Boosting

Dominates tabular AML. Consistent with Grinsztajn et al. (2022, NeurIPS): tree-based models outperform DL on structured data.

Method	Typical AUC	Speed	Key Advantage
XGBoost	0.95–0.99	Medium	Best overall tabular; built-in feature importance
LightGBM	0.95–0.98	Fastest	3–5× faster; high-cardinality cats
CatBoost	0.95–0.98	Slow	Native categoricals; ordered boosting
Stacking	0.96–0.99	Slowest	Best combined performance

Key findings:

- IEEE-CIS Fraud Detection: all top-10 used XGBoost/LightGBM. Winning AUC ∼0.966.
- CatBoost advantage disappears with manual target encoding (Oliveira et al., 2021).
- LightGBM within 0.5% AUC of XGBoost but 3–5× faster training—matters for production retraining.
- Lokanan et al. (2024, *J. Financial Crime*): RF (AUC 0.94) and XGBoost (0.95) match or beat LSTM (0.93) on tabular AML data.
- Stacking (RF + XGBoost + LightGBM + LR meta-learner) improves AUC by 0.5–1.5% over best single model.

3.2 Unsupervised Methods

Method	AUC	Strengths	Weaknesses
Isolation Forest	0.78–0.85	No labels; fast; scalable	Lower precision standalone
Extended IF	+1–2%	Better correlated features	–
LOF	0.72–0.80	Local anomalies	Doesn't scale; k-sensitive
One-Class SVM	0.70–0.80	Only needs normal class	Kernel choice; doesn't scale
DBSCAN	0.60–0.75	Structural outliers	Low precision
PCA	0.70–0.85	Fast; interpretable	Misses nonlinear

Comparative ranking (Hilal et al., 2022): IF > Autoencoder > LOF > OCSVM > DBSCAN.

Key insight. Unsupervised methods are best as *feature extractors*, not standalone detectors. Two-stage approach: IF anomaly scores as features for supervised model improves AUC by ∼3% (Jullum et al., 2020). This "anomaly score as feature" paradigm is now standard.

3.3 Hybrid Rule + ML

Industry standard. Pure ML replacement of rules is rare due to regulatory requirements:

- **Three-tier FRAML architecture** (2023): (1) hard rules for regulatory compliance, (2) soft rules for known typologies, (3) ML for emerging patterns.
- Rocha-Salazar et al. (2021): Rules generate alerts; RF re-scores. Alert volume −70%; SAR precision from 2% (rules) to 15% (hybrid).

- Soltani Delgosha and Hajiheydari (2021): Staged rule+ML: -40% false positives while maintaining recall.
- Regulators mandate explainable triggers $\Rightarrow$ rules for audit trail, ML for ranking.

4 Contextual Anomaly Detection

Core idea. A \$50K wire is normal for a corporate treasury but anomalous for a student account. Contextual anomaly detection models $P(\mathbf{y}|\mathbf{c})$—the expected *behavioral* features $\mathbf{y}$ given *context* features $\mathbf{c}$—rather than a global joint distribution $P(\mathbf{x})$. An observation is contextually anomalous if its behavior deviates from what is expected *given its context*, even if it appears normal globally.

This is a relatively thin literature compared to global anomaly detection, but it is the most operationally relevant framing for AML.

4.1 Foundational Methods

Method	Year	Key Idea	Venue
CAD	2007	Foundational: model $P(\mathbf{y}\|\mathbf{c})$ via conditional Gaussian mixture; anomaly = low conditional likelihood	IEEE TKDE
ROCOD	2016	Local (KNN-based) + global (regression) models with adaptive weighting by context density; 40× speedup	CIKM
QCAD	2023	Quantile regression forests for conditional quantiles; explainable via quantile intervals	DAMI
ConQuest	2024	Auto-discovers multiple useful contexts via genetic algorithm; MCAF multi-context scoring	IJDSA 2025

Song et al. (2007, IEEE TKDE). The foundational "conditional anomaly detection" paper. Splits features into context $\mathbf{c}$ and behavioral $\mathbf{y}$, models $P(\mathbf{y}|\mathbf{c})$ with conditional Gaussian mixtures. Anomaly score = negative log conditional likelihood. Showed contextual detection finds anomalies invisible to global methods.

ROCOD (Liang and Parthasarathy, 2016, CIKM). Addresses context sparsity: when few peers exist, the local model (KNN-based expected behavior) is unreliable. ROCOD blends a local peer model with a global regression model via adaptive weighting—local dominates in dense regions, global in sparse. 40× faster than prior contextual methods. The local model estimates expected behavior from contextual neighbors; the global model learns overall context→behavior dependencies.

QCAD (Li and van Leeuwen, 2023, *Data Mining and Knowledge Discovery* 37(6):2517–2563). Key advancement over ROCOD: uses *quantile regression forests* to model $P(\mathbf{y}|\mathbf{c})$ instead of the conditional mean. The quantile-based approach captures the full conditional distribution—wider quantile intervals indicate lower density and higher anomaly likelihood. Anomaly scores aggregated over behavioral features. Explainable: quantile intervals directly show why a point is anomalous ("value falls outside the 1st–99th percentile given this context"). Outperforms ROCOD on both accuracy and interpretability. Open-source: `github.com/ZhongLIFR/QCAD`.

ConQuest (Calikus, Nowaczyk, and Dikmen, 2024, *Int. J. Data Science and Analytics* 19:99–113). Solves the *context discovery* problem: which features should be context vs. behavioral? Uses a multi-objective genetic algorithm with three objectives: (1) maximize dependency between context and behavior, (2) minimize contextual redundancy, (3) maximize anomaly discrimination. Produces multiple useful contexts, each scored via a novel Multi-Context Anomaly Factor (MCAF). Outperforms SOTA on 25 datasets. Directly interpretable: each context provides a different "reason" for anomalousness.

4.2 Deep Contextual Methods

4.2.1 Conditional Wasserstein Autoencoder (CWAE)

King et al. (2025), "Contextual Learning for Anomaly Detection in Tabular Data" (arXiv:2509.09030, submitted to TMLR). This is the most directly relevant recent paper for deep contextual AD on tabular data.

Architecture. Splits features into context $\mathbf{c}$ and content/behavioral $\mathbf{y}$. Deterministic encoder $\mathbf{z} = E_\phi([\mathbf{y}; \mathbf{c}])$, decoder $\hat{\mathbf{y}} = D_\theta([\mathbf{z}; \mathbf{c}])$ reconstructs behavioral features only. Loss:

$$\mathcal{L} = \sum_{i=1}^{d} \mathrm{CE}(y_i, \hat{y}_i) + \lambda \cdot \mathrm{MMD}^2(P(\mathbf{z}), Q(\mathbf{z}|\mathbf{x}))$$

Uses MMD with multi-scale IMQ kernel instead of KL divergence $\Rightarrow$ deterministic encoder, no sampling noise, bounded gradients.

Context selection. Bilevel optimization: outer loop iterates over candidate context features; inner loop trains CWAE for 1 epoch; selects context minimizing joint validation loss $\mathcal{L}_{\mathrm{val}} = -\log P(\mathbf{y}, \mathbf{c})$. Leverages the finding that early-trained models proxy generalization quality.

Theoretical justification. Variance decomposition: $\mathrm{Var}(\mathbf{y}) = \mathbb{E}_{\mathbf{c}}[\mathrm{Var}(\mathbf{y}|\mathbf{c})] + \mathrm{Var}_{\mathbf{c}}(\mathbb{E}[\mathbf{y}|\mathbf{c}])$. A context-conditional detector only needs to model $\mathrm{Var}(\mathbf{y}|\mathbf{c})$, which is typically much smaller than total $\mathrm{Var}(\mathbf{y})$ $\Rightarrow$ higher SNR for anomaly detection.

Results on 8 benchmarks (Bank, Beth, Census, CMC, KDD, LANL, SF, Spotify):

Method	Avg AUC	Avg Rank	Type
CWAE (proposed)	**0.797**	**2.75**	Contextual deep
WAE (non-contextual)	0.760	4.25	Global deep
CVAE	0.745*	–	Contextual deep
CMDN	0.756*	–	Contextual deep
CADI	0.734*	–	Contextual tree
RCA	0.769	3.38	Global deep
DIF	0.758	3.50	Global tree
SLAD	0.695	5.88	Global deep

*Estimated from ablation tables; exact ordering varies by dataset.

Key finding: with oracle context selection, CWAE achieves avg AUC 0.839 (rank 1 on 6/8 datasets). The gap between automatic and oracle context selection highlights that *context feature choice* is the critical design decision.

CWAE vs. CVAE comparison. CWAE outperforms CVAE (+11.69% avg improvement over non-contextual baseline vs. CVAE's +6.21%). CWAE's deterministic encoder avoids CVAE's nu-

merical instability: CVAE requires $\log \sigma^2$ clamping to prevent divergence on skewed financial data, and Monte Carlo sampling adds inference noise.

4.2.2 Conditional VAE (CVAE) for Anomaly Detection

CVAE models $q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{c})$ with KL regularization. Applied to contextual AD by conditioning the generative model on context features. Three instability sources on financial data: (1) $\log \sigma^2 \to -\infty$ when encoder predicts near-zero variance, (2) $\log \sigma^2 \to +\infty$ causes overflow, (3) reparameterization amplifies noise when σ is extreme. The KL term also forces a "fuzzy" Gaussian latent space that may reconstruct anomalies too well.

Disentangled CVAE (IEEE 2024): Combines β-VAE, CVAE, and total correlation to disentangle context from anomaly signals. Addresses information loss while incorporating known sources of variation.

4.2.3 Other Deep Contextual Methods

- **CMDN** (Conditional Mixture Density Network): Neural network-parameterized Gaussian mixture for $P(\mathbf{y}|\mathbf{c})$. More flexible than single Gaussian but harder to train. +7.47% improvement over non-contextual (King et al., 2025).
- **CADI** (Contextual Anomaly Detection using Isolation Forest): Extends IF with density-aware splits conditioned on context. +4.18% improvement. Tree-based $\Rightarrow$ more interpretable.

4.3 Peer Group and Peer-Normalized Methods

The operational form of contextual AD in AML: compare each entity to its "peers" rather than the global population.

Bolton and Hand (2001). Foundational peer group analysis for fraud detection. Group entities by context features, flag deviations from peer behavior. The core idea: compute per-entity statistics relative to peer group mean and standard deviation.

Peer-normalized Z-score approach. For each entity i with context $\mathbf{c}_i$ and behavioral feature y_i:

$$z_i = \frac{y_i - \mu_{\text{peer}}(\mathbf{c}_i)}{\sigma_{\text{peer}}(\mathbf{c}_i)}$$

where μ_{peer} and σ_{peer} are estimated from context-similar entities (KNN with kernel weighting, or hard peer group assignment). This is effectively the simplest form of contextual AD.

Peer-Normalized Isolation Forest. Apply IF to peer-normalized features rather than raw features. This operationalizes contextual AD without changing the detector—only the feature representation changes. The normalization removes context-induced location/scale shifts, exposing within-context anomalies. Computationally cheap: no model retraining needed per context.

Uncertainty-Aware Contextual AD (arXiv:2507.04490, 2025). Extends the peer-normalized Z-score by treating it as a *random variable* with confidence intervals. Uses two heteroscedastic Gaussian process regression models—one for the conditional mean $\mathbb{E}[y|\mathbf{c}]$, one for the conditional variance $\text{Var}(y|\mathbf{c})$. Provides calibrated uncertainty: wide confidence intervals $\Rightarrow$ low reliability of anomaly assessment. Outperforms QCAD and ROCOD on benchmark datasets and a real-world cardiology application. Key advantage: enables adaptive, uncertainty-driven decision-making rather than hard thresholds.

4.4 When Does Context Help?

Not always. Contextual methods add value specifically when anomalies are *contextually unusual but globally normal*:

Anomaly Type	Example	Best Detector
Global	$10M single transfer (unusual for anyone)	Standard IF, XGBoost
Contextual	$50K wire from a student account (normal for corp.)	Peer-normalized IF, CWAE, QCAD
Structural	Fan-out → fan-in chain (smurfing)	GNN, FlowScope
Temporal	Dormant account suddenly active	LSTM, TGN

Key empirical finding: the magnitude of improvement from contextual methods depends on (1) the *type* of anomaly present and (2) the contamination rate. Under context-mismatch contamination, contextual methods can substantially outperform global methods (e.g., 39.8% vs. 20.8% precision at top-5% in one controlled experiment). Under global contamination, the advantage disappears or reverses.

Diagnostic heuristic. Low agreement between a global detector (IF) and a contextual detector (peer-normalized IF) at a given threshold can indicate the presence of contextual structure in the data. This can inform method selection before committing to a full pipeline.

4.5 KYC Integration in Practice

- **Customer 360 Profiles**: Dynamic integration of KYC data, transaction history, external datasets for trend/anomaly identification and peer group comparison.
- **HSBC Dynamic Risk Assessment**: ML models on transaction patterns + network behaviors + KYC data. Result: **60% alert reduction, 2–4× true positive increase**.
- **AI risk rating**: Incorporating behavioral and transactional factors into dynamic customer risk scores, enabling peer group comparisons.
- Customer segmentation (k-means, DBSCAN) enables contextual thresholding: entities far from cluster centroid get heightened scrutiny.

4.6 Entity Resolution

- **Knowledge graphs**: Connect people, companies, accounts, transactions. Trace ownership chains across borders/shells. Link duplicate records.
- Data-driven name matching: 90% more accurate than legacy fuzzy matching (handles transliterations, scripts, name variants).
- Graph-based entity resolution (Neo4j) for shared addresses, phones, IPs.
- Composite AI: entity resolution + graph analytics for proactive discovery.

4.7 Landscape Assessment

The honest summary. Contextual anomaly detection is a relatively thin literature:

- The core lineage is Song (2007) → ROCOD (2016) → QCAD (2023) → ConQuest (2024) → CWAE (2025).

- Most AML papers do global anomaly detection. Very few model $P(\mathbf{y}|\mathbf{c})$ explicitly.
- The deep contextual methods (CWAE, CVAE, CMDN) are recent (2024–2025) and largely untested on AML-specific data. King et al. (2025) benchmark on general tabular datasets (Bank, Census, KDD), not AML transaction data.
- Peer group analysis is widely used in industry AML (Bolton and Hand, 2001) but rarely published as formal ML research—it sits in the practitioner/vendor space.
- The context feature selection problem (which features are context vs. behavioral?) is largely unsolved. ConQuest's genetic algorithm and King et al.'s bilevel optimization are the only principled approaches.
- **Gap**: No published work combines CWAE/WAE-MMD with AML-specific evaluation on transaction monitoring data with realistic laundering typologies. Similarly, peer-normalized IF with systematic evaluation across injection strategies is understudied.

5 Large-Scale Entity Resolution

Entity resolution (ER)—also called record linkage, deduplication, or identity resolution—is the task of determining which records across (or within) datasets refer to the same real-world entity. At AML scale (100M+ clients across multiple systems), naive pairwise comparison is $O(n^2)$ and infeasible. This section covers blocking strategies, matching methods, and scalable architectures.

5.1 The Blocking Problem

Blocking is 10× more important than matching. Reducing candidates from n^2 to $0.01 \times n^2$ makes everything tractable. Modern blocking approaches:

Approach	Mechanism	Scale
Standard blocking	Hash on key attributes (name initial + DOB)	Simple; misses typos/variants
Sorted neighborhood	Sort by blocking key; slide window	Better recall; still key-dependent
LSH (Locality Sensitive Hashing)	Hash similar items to same bucket with high probability	Scalable; tunable recall/precision
ANN (Approximate Nearest Neighbor)	Embedding → FAISS/HNSW index → nearest neighbors	State-of-the-art; semantic similarity
Neural LSH	Learned hash functions via deep networks	Up to 6× faster index, 2× faster query (DET-LSH)

Neural LSH for Entity Blocking (Amazon, 2024, arXiv:2401.18064). Proposes training deep neural networks as hashing functions for complex metrics. Significantly outperforms traditional LSH on entity resolution benchmarks.

BlockingPy (2025, arXiv:2504.04266). Open-source package implementing ANN blocking via FAISS (exact KNN, LSH, HNSW). Designed for entity resolution at scale with imperfect data.

Hashed Dynamic Blocking. Applied to a 530M-row industrial dataset, detecting 68 billion candidate pairs in under three hours.

5.2 Matching Methods

5.2.1 Probabilistic: Fellegi-Sunter Model

The classical approach. For each record pair, compute match weights from field-level agreement/disagreement probabilities:

$$w_j = \log_2 \frac{P(\text{agree}_j|\text{match})}{P(\text{agree}_j|\text{non-match})}$$

Aggregate weights across fields; threshold for match/non-match/clerical review.

Splink (UK Ministry of Justice, open-source). The leading implementation of Fellegi-Sunter at scale. Key capabilities:

- Links 100M+ records using Spark or AWS Athena backends; 1M records on laptop in ∼1 minute.
- Unsupervised (no labeled training data): learns match/non-match weights via EM algorithm.
- Won Civil Service Awards 2025 (Innovation) and OpenUK Awards 2025.
- Used by Harvard Medical School on 8.1M records (American Journal of Epidemiology, 2025).
- Supports fuzzy string comparisons (Jaro-Winkler, Levenshtein), date comparisons, geographic distance.

5.2.2 Deep Learning: Transformer-Based Matching

The paradigm shift since 2020: use pre-trained language models (BERT, SentenceBERT, DistilBERT, RoBERTa) to encode record pairs into embeddings and classify match/non-match.

Ditto (Li et al., 2020). Landmark paper: BERT-based entity matching achieving SOTA, beating previous benchmarks by up to 29% F1. Key idea: serialize record pairs as text, fine-tune BERT as binary classifier. Introduced data augmentation for ER (attribute-level).

SentenceBERT for ER. SentenceBERT models rank first on most ER benchmarks because they natively produce embedding vectors suitable for similarity search. Used both for blocking (embed → ANN search) and matching.

EMBA (2024, EDBT). Multi-task learning of BERT for entity matching, improving over single-task fine-tuning.

Ensemble transformers (2024). Combining multiple transformer models for better entity matching through ensembling.

CSGAT (2025, *Scientific Reports*). Contextual Semantics Graph Attention Network extracting contextual information at token and attribute levels to generate semantically fused embeddings for entity matching.

5.2.3 LLM-Based Entity Resolution

The newest frontier (2024–2025):

- **Peeters and Bizer (2024)**: GPT-4 outperforms fine-tuned pre-trained language models by **40–68%** on unseen entity types in zero-shot settings. No training data required.
- **GPT-4o**: More cost-effective alternative; slightly lower performance than GPT-4.
- **In-context clustering ER** (2025, arXiv:2506.02509): Packs multiple records into sets for direct in-context clustering via LLM. Generalizes SOTA approaches with improved cost-effectiveness.

- **Balancing efficiency and quality** (Springer 2025): Explores optimal prompting strategies for LLM-based ER on structured data. "Selecting" strategy identified as most cost-effective.
- **Limitation**: Cost at 100M+ scale—each LLM call is expensive. LLMs best used for high-uncertainty pairs after initial blocking+probabilistic matching.

5.3 Scalable Architectures and Tools

Tool	Scale	Backend	Key Feature
Splink	100M+	Spark, Athena, DuckDB	Probabilistic (Fellegi-Sunter); unsupervised
Zingg	100M+	Spark, Snowflake, Databricks	ML-based blocking + matching; active learning
Senzing	Billions	In-memory graph	Real-time ER; entity-centric; used by banks
AWS ER	100M+	AWS Glue ML	Managed service; integrates with Zingg
Neo4j	100M+	Graph database	Graph-based ER; ownership chains

Zingg (open-source, AGPL v3.0). ML-based entity resolution for Spark, Snowflake, Databricks. Learns blocking model that indexes near-similar records—typical comparisons are 0.05–1% of the problem space. Supports both probabilistic (fuzzy) and deterministic matching. Active learning: human labels a small sample, model generalizes.

Senzing (commercial). In-memory, entity-centric ER engine. Processes billions of records in real-time. Used in financial crime, AML, and government applications. Key differentiator: maintains a live entity graph that updates incrementally as new records arrive—no batch reprocessing.

5.4 ER for AML: Specific Challenges

- **Name matching across scripts/cultures**: Transliterations (Arabic ↔ Latin), name ordering (family name first vs. last), aliases, nicknames. Data-driven name matching achieves 90% more accuracy than legacy fuzzy matching.
- **Shell companies and beneficial ownership**: Resolving entities behind layers of shell companies across jurisdictions. Knowledge graphs trace ownership chains; graph-based ER links shared addresses, phones, directors.
- **Real-time vs. batch**: AML requires both—batch for initial deduplication/linking across 100M+ records; real-time for new customer onboarding and ongoing monitoring.
- **Privacy constraints**: Cross-institution ER without sharing PII. Privacy-preserving record linkage (PPRL) uses encrypted/hashed tokens for comparison.
- **Scale progression**: <1M records: DuckDB/Python. 1–100M: Splink/Zingg on Spark. 100M+: Senzing, custom distributed architectures, or graph databases (Neo4j).

5.5 ER Performance Benchmarks

Leipzig benchmark datasets (Database Group Leipzig): Standard ER benchmarks with ground truth, including product, bibliographic, and geographic datasets at various scales.

Key results on benchmarks:

- BERT/Ditto (2020): +29% F1 over previous SOTA on structured ER benchmarks.

- SentenceBERT: Ranks 1st on most datasets; never below 4th.
- GPT-4 (zero-shot): +40–68% over fine-tuned PLMs on unseen entity types.
- Splink: Near-SOTA accuracy at 100M+ scale with zero training data.

The emerging architecture for 100M+ AML ER:

1. **Embedding**: SentenceBERT or domain-specific encoder → vector per record
2. **Blocking**: ANN index (FAISS/HNSW) → candidate pairs (0.01–1% of n^2)
3. **Matching**: Probabilistic (Splink) or fine-tuned transformer
4. **Uncertain pairs**: LLM (GPT-4) for high-ambiguity cases
5. **Graph consolidation**: Neo4j/knowledge graph for entity-level view

6 Feature Engineering

Feature engineering matters more than model choice. The gap between well-engineered XGBoost and poorly-engineered DL is larger than between model architectures.

Category	Examples	Impact
Velocity	Txn count/amount in 1h, 6h, 24h, 7d, 30d windows; unique counterparties; frequency acceleration; time since last txn	AUC +5–8% (Bahnsen et al., 2016)
Behavioral	Amount ratio to hist. mean; Z-score; new merchant/country; dormancy; day/time deviation	Most important category
Network/Graph	Degree; PageRank; clustering coeff.; fan-in/fan-out; cycle detection; community membership	F1 +7% even without GNNs (Alarab et al., 2020)
Aggregation	Per-customer stats (mean, std, skew); per-merchant fraud rate; per-country-pair amounts; per-channel stats	Foundation for behavioral deviation
Temporal	Day of week; hour; weekend/holiday; days since account open; seasonal patterns	Laundering timing patterns
Risk Indicators	FATF country risk; customer risk rating; PEP; industry risk; historical SAR count	Contextual conditioning

Two-stage feature architecture. Gaining popularity:

1. Unsupervised scoring (IF anomaly scores, PCA reconstruction errors, autoencoder loss, cluster distances)
2. Supervised classification on original features + Stage 1 scores

Addresses label scarcity. Jullum et al. (2020): ∼3% AUC improvement.

6.1 Handling Multicollinearity in Feature-Rich AML Pipelines

Feature engineering for AML produces dozens to hundreds of correlated features (e.g., txn count 24h ↔ txn count 7d; amount mean ↔ amount median). This multicollinearity does not degrade *prediction* in tree-based models but causes three practical problems: (1) unstable/misleading feature

importance, (2) inflated model complexity, and (3) attribution dilution across correlated features in post-hoc explanation methods—undermining explainability.

Detection.

- **Pearson / Spearman correlation matrix**: Drop one of any pair with $|r| > 0.90$–0.95. Simple, interpretable, but only captures pairwise linear relationships.
- **Variance Inflation Factor (VIF)**: Regresses each feature against all others. VIF $>$ 5–10 indicates problematic collinearity. More comprehensive than pairwise correlation but expensive at high dimensionality ($O(p^2 n)$).
- **Eigenvalue analysis**: Small eigenvalues of the feature covariance matrix indicate near-linear dependencies. Condition number > 30 signals severe multicollinearity.

Mitigation strategies.

Method	How It Works	When to Use
Correlation filter	Drop one of each highly correlated pair ($\|r\| > 0.9$)	Quick first pass; interpretable
VIF-based removal	Iteratively remove highest-VIF feature until all VIF < 5	Linear models, logistic regression
Variable clustering (VARCLUS)	Hierarchical clustering of features; keep one representative per cluster	Large feature sets; preserves coverage of feature space
PCA / kernel PCA	Project onto orthogonal components	When interpretability of individual features is not required
Mutual information (mRMR)	Select features maximizing relevance to target while minimizing redundancy among selected features	Supervised setting with labels; captures nonlinear dependencies
LASSO / Elastic Net	L_1 penalty drives correlated feature coefficients to zero	Embedded selection; regularization path shows stability
Recursive Feature Elimination (RFE)	Iteratively remove least-important features using model importance	Works with any model; computationally expensive
Attribution-based selection	Cluster features by explanation-value correlation; keep representative of each cluster	Post-hoc; directly addresses attribution dilution

Practical notes for AML feature engineering:

- **Tree models (XGBoost/LightGBM) are robust to multicollinearity for prediction**—model accuracy is minimally affected. However, feature importance and post-hoc explanations become unreliable when features are correlated (importance "splits" across redundant features). Since explainability is required for FINTRAC, this must be addressed.
- **Recommended pipeline**: (1) Correlation filter ($|r| > 0.9$) as first pass, (2) group remaining features by semantic category (velocity, behavioral, network, risk), (3) within each group, apply VIF or mRMR to select non-redundant representatives, (4) validate with attribution clustering post-training.
- **Domain-informed grouping**: Rather than purely statistical removal, group by business meaning. E.g., keep `txn_count_24h` and `txn_count_30d` (different time horizons) but drop `txn_count_23h` (redundant with 24h). Domain knowledge prevents removing features that are statistically cor-

related but semantically distinct.

- **Interaction analysis**: For pairs that survive filtering, interaction values (e.g., SHAP ϕ_{ij} or permutation-based methods) can reveal whether both features contribute *independently* to predictions. If interaction is near-zero and main effects are similar, one can be dropped.
- **Stability selection** (Meinshausen & Bühlmann, 2010): Run feature selection (LASSO, RFE) on random subsamples; keep features selected in $> 60\%$ of runs. Robust to collinearity-induced instability.

7 Datasets and Evaluation

7.1 Public Datasets

Dataset	Type	Size	Imbal.	Notes
Elliptic	Real BTC	203K txns	2%	Most-used; 49 timesteps; 166 features
Elliptic++	Real BTC	204K+	2%	+address-level graph; subgraph labels
AMLSim (IBM)	Synthetic	Config.	Config.	Open-source; controllable patterns
PaySim	Synthetic	6.3M txns	0.13%	Mobile money; network structure
IBM AML Synth.	Synthetic	180M+	Config.	Large-scale multi-bank
IEEE-CIS	Real CC	590K	3.5%	Kaggle competition; rich features
European CC	Real CC	284K	0.17%	PCA-anonymized

Synthetic data. Altman et al. (2023, *Nature Scientific Data* / NeurIPS 2023): large-scale synthetic AML benchmark. Market projected \$310–576M (2024) → \$1.8–16.7B by 2030. By 2025, 75% of large banks use synthetic data for AI. Tools: CTGAN, Copula GAN, TVAE—96–99% utility equivalence.

7.2 Class Imbalance

AML: 0.1–2% positive rate. Approaches:

- **SMOTE**: Most widely used. RF+SMOTE: F1 from 0.72 → 0.87 (Al-Hashedi and Magalingam, 2021).
- **SMOTE-GAN hybrids**: SMOTified-GAN, etc.
- **Cost-sensitive learning**: Class weights, focal loss.
- **GraphSMOTE**: Graph-aware oversampling.
- **AMLNet** (2025): SMOTE + RandomUnderSampler + multi-agent framework.

7.3 Metrics

AUC alone is insufficient for AML:

- **F1 (illicit)**: More discriminating on imbalanced data. Ranges 0.47 (RF) to 0.88 (PC-GNN).
- **Precision@top-k**: How many top-k alerts are true? Maps to analyst workload.
- **Alert-level precision**: % of alerts → SARs. Rules: ∼2%; hybrid ML: ∼15%.
- **FP reduction**: Primary industry KPI. ML consistently 40–60% over rules.

- **PSI**: Monitors concept drift; triggers retraining.

7.4 Concept Drift

Launderers adapt ⇒ model degradation:

- Deprez et al. (2025): Continual graph learning review—replay, regularization, architecture strategies.
- Cross-regional continual learning with knowledge transfer (TKDE 2024).
- PSI-based drift detection standard in production.
- Traditional models suffer catastrophic forgetting on fine-tuning.

8 Emerging Paradigms

8.1 Federated Learning

Banks can't share data but laundering spans institutions:

- **FAML**: Multi-bank collaborative training without data sharing. Up to **30% higher accuracy** vs. isolated.
- **FedGraph-VASP** (2026): Federated graph learning + post-quantum crypto for cross-institutional AML.
- One implementation on PaySim: 0.999 accuracy, 0.1% false alarm rate.
- **FHE + GNNs**: Fully homomorphic encryption for privacy-preserving graph ML.
- **IBM Research**: Privacy-preserving federated learning for financial crime.
- **Federated Graph Anomaly Detection via Disentangled Representations** (TheWebConf 2025).

8.2 Explainability

Regulatory non-negotiable (FinCEN, FCA, EBA, FATF):

- **SHAP/TreeSHAP**: Widely used. Jullum et al. (2020) reported analysts 30% faster with SHAP explanations. However, SHAP values can be unstable with correlated features (common in AML), and computational cost grows with feature count. Should be used alongside—not as a replacement for—domain expertise.
- **LIME**: An alternative to SHAP for local explanations, though less stable on correlated tabular features. Useful for quick per-alert explanations but sensitive to perturbation strategy.
- **SEFraud** (KDD 2024): Self-explainable fraud via interpretative mask learning.
- Counterfactual explanations ("this alert would not have fired if...") demanded by regulators.
- Global surrogates (decision trees/GAMs approximating black-box) for regulatory documentation.
- EU AML Directive (2020), FinCEN guidance (2021): mandate "understandable" automated detection.

8.3 Agentic AI

Most recent frontier (2025–2026):

- **Nasdaq Verafin**: Agentic AI Workforce automating low-value, high-volume compliance. Up to 50% AML workload automation.
- **Co-Investigator AI** (2025): Agentic SAR generation—faster and more accurate.

- McKinsey (2025): Agentic AI automates EDD reviews (adverse media, historic activity, escalation).
- Agent-based KYC: onboarding from weeks → hours.
- Human-in-the-loop: AI handles lower-risk; humans retain high-level decisions.

8.4 Adversarial Robustness

- CARE-GNN: Anti-camouflage (fraudsters connecting to benign nodes).
- RoBCtrl (2025): RL-based attacks on GNN bot detectors—highlights deployed system vulnerability.
- Adversarially robust temporal graph contrastive learning (2025).
- Graph adversarial training (2024) for GNN robustness.

9 Decision Framework

Scenario	Data	Labels	Recommended Approach
Tabular, labeled	Tabular	Yes	XGBoost/LightGBM + post-hoc explanations
Tabular, no labels	Tabular	No	IF + two-stage
Graph, labeled	Network	Yes	GraphSAGE/GAT + temporal
Graph, no labels	Network	No	GAE or self-supervised GNN
Multi-institution	Any	Mixed	Federated learning (FAML)
Real-time	Stream	Any	TGN / streaming GNN
Regulatory focus	Any	Any	Hybrid rule+ML + explainability layer
Few labels	Any	Few	Self-supervised + fine-tuning

10 Industry Adoption

Organization	Method	Scale	Year
HSBC	Dynamic Risk Assessment (ML+KYC)	Banking	2024
Ant Group	Heterogeneous GNN	Billions of nodes	2020+
eBay	xFraud (Het-GNN)	Billions of txns	2022
PayPal	GNN + streaming	Real-time	2021+
Feedzai	Graph-enhanced ML	Banking AML	2022+
Elliptic	AI on Bitcoin	200M txns	2024
Nasdaq Verafin	Agentic AI	Banking AML	2025

SAS/KPMG/ACAMS 2025 survey (850 ACAMS members): 18% have AI/ML in production; 18% piloting; 25% plan in 12–18 months; 40% no plans. Top priority: FP reduction (38%). ML top tech choice (58%). Top barrier: lack of regulatory imperative (37%, overtook budget). Regulator enthusiasm declining (51% say regulators encourage, down 15pts from 2021).

11 Supervised Learning with STR-Labeled Data

When dispositioned STR (Suspicious Transaction Report) data is available, supervised learning becomes viable and is generally the highest-performing approach.

11.1 Training Data Construction

- **Label source**: Filed STRs = positive class; reviewed-and-cleared alerts = negative class. Raw non-alerted transactions are *not* true negatives (unknown positives problem).
- **Recommended window**: 6–24 months of dispositioned cases. Too short = insufficient positive examples; too long = concept drift contaminates the training set.
- **Label quality**: STR filing decisions vary by analyst. Consensus labels (multiple reviewers) or adjudicated labels (senior review) improve training signal. Label noise is the single largest risk to supervised AML models.
- **Class imbalance**: STR filing rates are typically 2–5% of alerts, which are themselves <1% of transactions. Effective mitigation: SMOTE, cost-sensitive learning ($w_{\text{pos}} = N_{\text{neg}}/N_{\text{pos}}$), or focal loss. GraphSMOTE extends oversampling to graph-structured data.

11.2 Supervised Model Performance

Gradient-boosted trees dominate supervised AML:

- XGBoost on tabular AML features: **AUROC 0.961** reported on production-scale labeled data (Jullum et al., 2020; DNB Bank study).
- LightGBM: comparable AUROC with 2–3× faster training; preferred at scale.
- Random Forest: strong baseline (AUROC ∼0.94) but less competitive than boosting.
- Deep learning (DNN, LSTM): marginal gains over XGBoost on tabular data; justified only when sequential or graph structure is exploited.

False positive reduction is the primary value proposition of supervised models over rule-based systems. Literature consistently reports **60–80% FP reduction** while maintaining or improving recall:

- Deloitte/SAS (2024): 60% FP reduction with supervised ML overlay on existing rules.
- Guidehouse (2023): ML models reduce alert volumes by 70% with no loss in STR filing rate.
- FinCEN (June 2024 proposed rule): explicitly endorses ML for "greater precision in assessing customer risk" and "reducing false positives."

11.3 Semi-Supervised Extensions

When labeled data is limited (common in early deployment):

- **Two-stage pipeline**: (1) unsupervised anomaly scores (IF, autoencoder, LOF) as features, (2) supervised classifier on the enriched feature set. This is the most common production architecture.
- **Self-training / pseudo-labeling**: Train initial model on labeled subset, predict on unlabeled, add high-confidence predictions to training set. Iterate.
- **Positive-unlabeled (PU) learning**: Treats non-STR transactions as unlabeled (not negative). Appropriate given the unknown-positives problem. Methods: Elkan & Noto (2008), unbiased PU risk estimator (Kiryo et al., 2017).
- **Contrastive pre-training**: Self-supervised on full transaction corpus, then fine-tune on STR labels. FLAG (KDD 2025), TREASURE (2025) demonstrate this for financial data.

12 Complex Behavior Detection

FINTRAC and international regulators increasingly focus on *efficacy*—the ability to detect complex, multi-step laundering typologies, not just individual suspicious transactions.

12.1 Laundering Typologies

- **Structuring (smurfing)**: Breaking large amounts into sub-threshold deposits across accounts/branches. Rule-based detection (aggregation thresholds) is well-established but easily evaded by varying amounts.
- **Layering**: Moving funds through multiple intermediary accounts/entities to obscure the trail. Requires multi-hop graph analysis.
- **Mule networks**: Recruiting third-party accounts to receive and forward funds. Detected via fan-in/fan-out patterns and behavioral anomalies.
- **Trade-based laundering (TBML)**: Over/under-invoicing in international trade. Requires trade data integration and price anomaly detection.
- **Nested/shell structures**: Using corporate hierarchies to obscure beneficial ownership. Entity resolution is critical here.

12.2 Detection Approaches by Typology

Typology	Best Approach	Key Reference
Structuring	Temporal aggregation + supervised	XGBoost with rolling-window features
Smurfing	Subgraph mining	GARG-AML (2025): second-order neighborhood scoring; SMoTeF: temporal-constrained flow
Layering	Multi-hop GNN / path analysis	FlowScope (AAAI 2020): dense subgraph on flow networks
Mule networks	Community detection + fan patterns	Heterogeneous GNN with role-aware embeddings
TBML	Price anomaly + cross-dataset	Trade price deviation models + customs data

12.3 Multi-Typology Supervised Models

With STR labels, a single model can learn multiple typologies simultaneously:

- **Multi-label classification**: Each STR is tagged with typology code(s). Train XGBoost/LightGBM with multi-label targets. Enables typology-specific explanations.
- **Hybrid LSTM-GraphSAGE**: Captures both sequential transaction patterns (LSTM) and network structure (GraphSAGE). Reported **95.4% accuracy** on simulated multi-typology data.
- **Hierarchical detection**: Level 1 = anomaly score (unsupervised), Level 2 = typology classifier (supervised), Level 3 = case-level aggregation. Each level adds specificity.

13 FINTRAC Regulatory Alignment

FINTRAC's evolving stance creates both opportunity and obligation for ML-based AML systems:

- **Efficacy mandate**: New violation for failure to ensure compliance programs are "reasonably designed, risk-based and effective" (2025). Shifts focus from checkbox compliance to demonstrable detection capability.
- **Enforcement escalation**: 20 AMPs in 2025 alone, exceeding $200M total—the largest in FINTRAC history. Inadequate transaction monitoring is a primary target.
- **FinCEN endorsement of ML**: June 2024 proposed rule explicitly endorses AI/ML for "greater precision in assessing customer risk" and "reducing false positives." Canadian regulators are expected to follow.
- **OSFI E-23 Model Risk Management** (effective May 2027): Will require model validation, ongoing monitoring, and documentation for all models used in risk management—including AML. Architecture must be designed with validation in mind from day one.
- **Explainability requirement**: Both FINTRAC and OSFI require that automated decisions be "understandable." Post-hoc explanation methods (e.g., SHAP, LIME, surrogate models) can help satisfy this, though no single method is universally sufficient. Black-box models (deep GNNs, LLMs) require additional justification.

Note: Some citation details (exact page/volume numbers) should be verified against original sources before formal publication. Performance metrics reflect literature-reported values under specific experimental configurations.